

Student Project

Environmental Economics — BSc AIDAMS

T1 2026–2027

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1. Objective

You will work with real data on a question in climate and environmental economics, and present your findings to the class.

The purpose is not to summarise what others have written. It is to **obtain a dataset, understand what it actually measures, and use it to answer a specific question**. A clear, honest analysis of a simple and clear question is worth more than an ambitious claim that your analysis cannot support.

Every track below is built on a public dataset that should be freely accessible. You may add or prefer other sources if you wish.

2. Organisation

- **Groups of 5.** Ten groups, one per track. Tracks are allocated; no two groups work on the same one.
- **Paper:** maximum 10 pages, submitted by **Monday 2 November 2026** (session 6).
- **Presentation: 10 minutes** plus 3 minutes of questions, on **Monday 9 November 2026** (session 7).

In the presentation, two members speak and the others answer questions. Everyone is expected to know the whole project.

Start with the data. Every track names a source that should be freely available, but downloading it, working out what it contains and assembling it into something usable always takes longer than expected — and some sources are reached through a web interface or a form rather than a direct download. Get the data in hand in the first days: a problem found early is a problem you can still solve.

3. What every paper must contain

The same structure applies to all ten tracks. **Maximum length: 10 pages**, including figures and tables, excluding references and the statement on the use of AI. The lengths below are indicative and need not add up to ten.

1. **The question** *about 1 page*
What you are asking, why it matters, and for whom. State the question precisely enough that a reader knows what would count as an answer.
2. **Description of the data** *about 1–2 pages*
Cover: the source and who produces it; geographical and time coverage; frequency; units of measurement; how the variables are constructed; missing observations and how you handled them; known limitations of the dataset. If you merge two sources, explain how you matched them and what you lost in the process.

3. Summary statistics*about 1 page*

In the form of a table, for example: means, dispersion, range, number of observations, by the dimension that matters for your question.

4. Figures*about 3 pages*

Each figure must answer part of your question and must stand on its own: axes labelled with their units, coverage and period stated, source given. A reader who looks only at the figure and the sentence below it should understand both what is shown and what to conclude from it, without reading the surrounding text. A figure from which nothing relevant to the question can be concluded does not belong in the paper.

5. Econometrics*optional, about 2 pages*

By econometrics we mean going beyond describing the data to estimating a relationship: writing down an equation, estimating its coefficients, and reporting how precise those estimates are. A regression of one variable on one or more others is the usual case; a fitted trend reported with its standard error already counts.

If you include it, name the estimation method — ordinary least squares, a panel regression with fixed effects, local projections, and so on — write the equation, say what every variable is and in what units, report standard errors next to the coefficients, and interpret the size of the coefficient in the units of the problem — not only its sign and whether it is significant.

6. What we can and cannot conclude*about 1 page*

Required for everyone, including those who skip section 5. State the method you used, and whether your result is a description, a correlation, a prediction, or a causal effect, and why. If it is a correlation, say so plainly and explain what would be needed to make it causal.

On correlation and causality. Most of these tracks are descriptive questions, and answering them well is a complete piece of work. You lose marks for claiming a causal effect that your design does not support: a well-executed description, honestly labelled, scores higher than a causal claim that does not hold.

4. Use of AI and reproducibility

You may use AI tools. At the end of the paper, state in a short paragraph which tools you used and for what — writing, code, literature search, or anything else. Declared use is not penalised; undeclared use is.

Whatever you submit is yours: you are responsible for every number, every figure and every statement in it, and you must be able to explain any of them. A result you cannot explain will be treated as one you did not obtain.

Submit your code with the paper. Your code needs to reproduce all your results, starting from the raw data file as you downloaded it. If the dataset is small, include it; if it is large, include the script that retrieves it.

Every number must be traceable. For each dataset, give the producer, the dataset name, the version or release date, and the date you accessed it. Figures quoted without a traceable source are treated as unsupported — and note that language models produce plausible numbers that are sometimes wrong, so anything you did not read off the data yourself needs checking against it.

In the Q&A after your presentation, you may be asked about specific choices in your data and figures.

5. Tracks

Climate, weather and satellite data

Track 1 — Temperature and precipitation records

Question What do the observed records show for a set of countries? How have average temperature and precipitation changed over the period covered by the data, how much do they vary from one year to the next, and where has the change been largest? And do the years that stand out in the climate data also stand out in the economy?

Data The Copernicus Interactive Climate Atlas (<https://atlas.climate.copernicus.eu/>) covers both variables. You select one or more countries from its predefined regions, choose temperature or precipitation, and download the resulting time series as a CSV file — so the data arrive already aggregated to country level, with no gridded files to process and no account to create. The gridded data underneath the Atlas are also published in the Copernicus Climate Data Store if you ever need finer spatial detail, but that route requires an ECMWF account and returns files you would have to aggregate yourself. For the last part of the question, the World Bank’s World Development Indicators give a country-year panel of economic variables — growth of GDP per capita, the share of agriculture in value added, and many others — free and without a key (<https://api.worldbank.org/v2/>).

What to do (1) Plot temperature and precipitation over the full available record for a set of countries chosen across different regions. (2) Compute a *monthly anomaly*: for each calendar month, subtract the average of that same month over a baseline period that you choose and state — 1961–1990 is a common one. Without this step the seasonal cycle dominates everything and one month cannot be compared with another. (3) Describe each series: how much has the average changed between the start and the end of the record, and how large is the year-to-year variation compared with that change? Then compare countries — where has temperature moved most, and does precipitation move in the same places or in different ones? (4) Finally take one of two directions. Either bring in an economic variable of your choice from the World Development Indicators and put it next to your climate series — were the years that were unusually warm or unusually dry also unusual for the economy? Or stay inside the climate data and look at the shape of the change: using the monthly anomalies from step (2), is the change spread evenly across the year, or are some seasons moving faster than others?

Note Check whether the series you download are absolute values or already expressed as anomalies against some baseline — if they are anomalies, you are adopting that baseline rather than choosing one, and should say which it is.

Track 2 — Extreme events: a shift or a change in shape?

Question Are extreme events becoming more frequent or more intense — and is that because the whole distribution has shifted, or because its tail has changed shape? The mean of a climate variable can rise while its extremes behave exactly as before, and the tail can grow heavier while the mean barely moves.

Data The same Copernicus Interactive Climate Atlas used in Track 1 also carries extreme-event indices: 35 variables and indices in total, grouped into heat and cold, wet and dry, drought, wind and radiation, and snow and ice — among them frost days, heating degree days and standardised drought indices such as SPI and SPEI. As in Track 1 you select countries, choose an index and download the series as CSV, already aggregated to country level (<https://atlas.climate.copernicus.eu/>). The gridded data underneath sit in the Copernicus Climate Data Store for anyone who needs finer resolution, at the cost of an ECMWF account and of aggregating the files yourself.

What to do (1) Choose one or two indices and a set of countries, and say why. (2) Split the record into an early period and a recent one, and compare the two *distributions* rather than only

their averages: plot them together, and report the mean, the spread and the upper percentiles of each. (3) Say which of the two stories your data tell — a distribution that has moved as a whole, or one whose tail has changed shape — and by how much. (4) Then vary the threshold that defines the extreme and redo the comparison: how much of your answer depends on where the line is drawn?

Note An index is a definition, not a fact: what counts as a heatwave or a dry spell depends on the threshold chosen, and the Atlas will offer you several. State the index you used and how it is defined. Note also that extremes are rare by construction, so the upper tail rests on few observations — say how many, and be careful about conclusions drawn from a handful of events.

Track 3 — Satellite estimates against declared inventories

Question Do satellite-derived emission estimates agree with official inventories? Where do they diverge, and is the divergence systematic by country or by sector?

Data Climate TRACE estimates emissions from satellite observations and other measurements for individual facilities, and publishes the totals by country and sector, free and under CC 4.0, as downloads or through a public API (<https://climatetrace.org/data>). EDGAR, the European Commission’s inventory, builds its estimates the opposite way: from reported activity data multiplied by emission factors (<https://edgar.jrc.ec.europa.eu/>). The two are constructed by completely different methods, which is what makes the comparison interesting.

What to do (1) Line the two up for a set of countries, a common year and a matched sector classification. (2) Compute the difference, both in absolute terms and as a share of the level. (3) Ask whether the differences are systematic: by sector, by country, by gas.

Note No image processing is required — Climate TRACE publishes its estimates in tabular form. The sector classifications differ between the two sources, so the matching is your decision and belongs in the data description. A difference is not an error by either side: they measure different things in different ways, and your job is to say what each one includes.

Track 4 — Working with climate scenarios

Question Within a single NGFS scenario, how much do the underlying integrated assessment models disagree? What does that dispersion mean for someone using these numbers in a stress test?

Data The NGFS Scenario Explorer reports the same variables for several integrated assessment models within each scenario. The models are built by different teams with different assumptions about how the economy responds, so they can produce different numbers for the same scenario (<https://data.ece.iiasa.ac.at/ngfs/>). Access is free and needs no registration: on the Explorer, click *Guest login* to reach the download page, where the data come as CSV or xlsx. A REST API and the Python package `pyam` give the same data programmatically.

What to do (1) Choose one variable — GDP loss or the carbon price, for instance — and one scenario. (2) Plot the path produced by every model within that scenario. (3) Compare the spread between models inside a scenario with the distance between the scenario averages, and state what range of uncertainty a user of these numbers should carry.

Note The spread is not measurement error. It is genuine disagreement about how the economy responds, and quantifying it is the point — not judging which model is right.

Economics and finance

Track 5 — Green innovation and technology costs

Question Are the energy technologies that attract the most patents and R&D also the ones whose unit cost (per same production of energy) falls fastest? Where do innovation and cost decline move together, and where do they diverge?

Data IRENA publishes, for every renewable technology and every year, the average cost of producing one megawatt-hour of electricity — the levelised cost of electricity, or LCOE — together with the cost of installing one kilowatt of capacity and the share of the time that capacity actually runs (<https://www.irena.org/Data/View-data-by-topic/Costs/Global-Trends>). The OECD Data Explorer publishes counts of patents filed in environment-related technologies, divided into more than 80 technology classes (<https://www.oecd.org/en/data/indicators/patents-on-n-environment-technologies.html>). Both are free, with one caveat: IRENA’s project-level cost database is reserved to members of its Costing Alliance, so what is public — and what you will use — are the published annual averages by technology. Installed capacity is fully open through IRENASTAT, which offers a query tool and a bulk download by country, technology and year from 2000 (<https://pxweb.irena.org/pxweb/en/IRENASTAT/>).

What to do (1) For each technology, plot the cost of one megawatt-hour against the total capacity installed worldwide up to that year, with both axes on a logarithmic scale. If cost falls by a constant percentage every time cumulative capacity doubles, the points line up along a straight line; its slope is what is called the *learning rate*. A steep line means the technology gets cheaper quickly as it is deployed, a flat one that it does not. (2) Separately, plot the number of patents filed per year in the matching technology class. (3) Then put the two together and ask whether the technologies with the steepest cost decline are also the ones with the most patenting.

Note The two sources name technologies differently, so you will have to decide which patent class corresponds to which IRENA category — for instance which class covers photovoltaics. You might decide to restrict to the technologies where the correspondence is clearest, and write down the choices you made and why in your data description.

Track 6 — Carbon pricing and emission reductions

Question How have emissions changed since the EU ETS began, sector by sector? Which sectors have cut them most, which have not moved — and where the intensity of an industry has fallen, is that because it emits less or because it produces more?

Data Eurostat publishes, for each of 64 industries classified by NACE and each year from 1995 to 2024, the greenhouse gases emitted per unit of value added and per unit of output — the *emissions intensity* of that industry. Value added is what an industry adds to the value of the inputs it buys, so the intensity says how much emission is needed to generate one euro of it. The dataset is `env_ac_aeint_r2` (https://ec.europa.eu/eurostat/databrowser/view/env_ac_aeint_r2/default/table?lang=en). Two companion datasets are needed for the last part of the question: `env_ac_ainah_r2` gives the emissions themselves, in tonnes, for the same industries, and `nama_10_a64` gives gross value added by industry. Value added could also be recovered by dividing emissions by intensity, but taking it directly is cleaner and lets you check that the three sources agree.

What to do (1) Plot emissions intensity over time for a selection of sectors, including some covered by the ETS and some that are not. (2) For each sector, compute the percentage change since the ETS began in 2005 and rank them: which industries cut the most, which did not move. (3) For the sectors that improved most, separate the two possible reasons: put absolute emissions and value added side by side and say whether intensity fell because emissions came down or because output grew faster than them.

Note Not every sector is covered by the ETS: check which ones are before comparing, because a fall in an uncovered sector cannot be attributed specifically to the carbon price. And a falling intensity is not the same as falling emissions — that distinction is the point of step (3).

Track 7 — Stranded assets

Question How much fossil capacity would have to retire before the end of its technical life along the different transition pathways, and where is that concentrated — by country, technology and plant age?

Data The NGFS Scenario Explorer gives, for each scenario (Net Zero 2050, Delayed Transition, Current Policies and others) and each underlying model, projected paths to 2050 for electricity generation by fuel, emissions and the carbon price. Access is by guest login, with CSV and xlsx downloads, a REST API and the Python package `pyam` (<https://data.ece.iiasa.ac.at/ngfs/>); on the Explorer, click *Guest login* to reach the downloads — free, with no registration. For the existing stock, the Global Energy Monitor trackers list power plants worldwide with their capacity, country and commissioning year (<https://globalenergymonitor.org/projects/global-coal-plant-tracker/download-data/>). That data is free and released under CC BY 4.0, but it is not a direct download: you fill in a short form with an email address and receive the file.

What to do (1) From the plant inventory, work out how much coal and gas capacity each country has and when it was built; assume a technical lifetime and derive the year each plant would retire naturally. (2) From the scenarios, take the path of fossil generation each one implies. (3) Compare the two: how much capacity would have to close earlier than its natural retirement date, in which countries, and at what age.

Note The technical lifetime of a plant is not a column in the dataset, but it need not be invented: the tracker also records units that have already closed, with both their commissioning and their retirement year, so the distribution of realised lifetimes can be estimated from the data themselves and compared with published figures. Say which lifetime you use and where it comes from, and show how much the answer moves if you change it.

Track 8 — Carbon policy shocks and sovereign risk in the euro area

Question How does a carbon-policy shock transmit to the euro-area economy — energy prices, inflation, activity — and does it widen the sovereign spread?

Data Känzig measures how much the price of carbon futures moved in the narrow window around each EU ETS regulatory announcement, which isolates news about carbon policy from the state of the economy. Take the monthly series and sum it to quarters (<https://github.com/dkaenzig/carbonpolicyshocks>, CC BY 4.0). Build the euro-area macro panel from Eurostat — HICP headline and its energy component, industrial production, unemployment — and take the policy rate and ten-year government bond yields from the ECB Data Portal (<https://data-api.ecb.europa.eu/service/data/>). The sovereign spread is a country's ten-year yield minus the German Bund. The sample runs 1999Q1–2019Q4, which is 84 quarters.

What to do (1) Build the quarterly dataset: aggregate the shock, assemble the macro series, construct the spread, and check the sample lines up. (2) Plot the shock series and describe it — how often the events occur, how large they are, and in which years the big ones fall. (3) Put the sovereign spread and energy inflation alongside it. Mark the quarters with the largest shocks and describe what happens in the quarters that follow: compare the average path after a large shock with the average across all quarters. Do this for one high-debt and one low-debt country and compare them. (4) If you take the econometrics option, estimate the same responses with local projections or a proxy-SVAR over about twenty quarters, with confidence bands. Both methods are standard on this shock and widely documented, so there is plenty of published material to work from. Then say whether the estimated picture agrees with what you saw in step (3).

Note Frequency is not a detail here: the shock is measured monthly but the macro series are quarterly, and the aggregation is what makes the two comparable. If you estimate, compute your own first stage and report it — at monthly frequency the instrument is weak, and if yours is weak say so rather than reading the response anyway.

Track 9 — Insurance and the protection gap

Question How has the gap between economic losses and insured losses moved over time and across countries? Which perils are least covered, and who bears the uninsured share?

Data The EIOPA dashboard reports, for 30 European countries, the economic losses caused by natural catastrophes and the share of those losses that was insured, from 1980 to 2024, broken down by peril — flood, windstorm, earthquake, wildfire, drought, hail. The difference between the two is the *protection gap*. The dashboard, its technical documentation and the input data are all downloadable (https://www.eiopa.europa.eu/tools-and-data/dashboard-insurance-protection-gap-natural-catastrophes_en). Swiss Re Institute *sigma* reports give the same picture at world level.

What to do (1) Compute the gap as the share of economic losses left uninsured, by country and by peril. (2) Plot how it has moved over time. (3) Compare countries and perils to find which combinations are least covered, and relate that to how exposed each country is.

Note Catastrophe losses are extremely uneven: one large event dominates a whole year. Work with multi-year averages, or show the distribution rather than single years.

Track 10 — Banking sector exposure to climate risk

Question How is the exposure of European banks distributed between transition risk — the emissions financed by their portfolios — and physical risk? Which countries and which portfolios are most exposed, and to which hazards?

Data The ECB, with the national central banks, publishes three families of indicators. *Financed emissions* attribute to a bank the emissions of the companies it lends to or holds securities in, in proportion to its stake. *Physical risk* indicators measure how exposed a portfolio is to river and coastal flooding, wildfire, landslide, subsidence, windstorm and water stress. *Sustainable finance* indicators cover green bond issuance and holdings. The first two are downloadable as compressed CSV, the third from the ECB Data Portal; last updated November 2025 (<https://www.ecb.europa.eu/stats/all-key-statistics/horizontal-indicators/sustainability-indicators/html/index.en.html>).

What to do (1) Rank euro-area countries by financed emissions per unit of exposure. (2) Do the same for physical risk, hazard by hazard. (3) Compare the two rankings: do transition risk and physical risk fall on the same countries, or on different ones?

Note The ECB labels the emissions and physical risk indicators *analytical* rather than official statistics, and says they should be used with care. Report that in your data description. Do not rely on the Green Asset Ratio from banks' Pillar 3 disclosures either: those templates are suspended and deferred to 2027.

6. Sources

Cite at least three policy reports or research papers — IPCC, World Bank, IMF, OECD, NGFS, ECB, or academic journals. The course bibliography is the natural place to start. Use a consistent reference style.

Cite the data as carefully as the literature: producer, dataset name, version or release date, and date of access.

7. Assessment

- **Description of the data.** Accuracy and honesty about what the data measures, its coverage and its limits.

- **Figures.** Whether each one can be read on its own and supports a clear conclusion.
- **Reasoning.** Use of economic argument; whether the interpretation follows from what was actually estimated.
- **Correlation and causality.** Whether the strength of the claim matches the design.
- **Critical analysis.** Not a summary: a position, argued.
- **Clarity** of paper and presentation.

The project counts for 40% of the final grade.